

Homework 6

STAT 300 (Fall 2024)

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Clearly label your answers to each question and each sub-question. Your answers MUST be uploaded to Canvas as a <HWx_Yourfirstname.nb.html> file by the deadline.

- (1) Certain statistics course include a midterm exam (out of 100) and a project (out of 30). Below least squares model was obtained using the data from 24 students in the class.

$$\widehat{Final} = 11.0 + 0.53Midterm + 1.20Projet$$

- (a) What would this tell you about a student who got perfect score on the midterm and the project?
 - (b) Leo got 87 on the midterm and 21 on the project, and 80 on the final exam. Calculate the residual and write a sentence explaining what this means in Leo's case.
 - (c) Does the fitted model above suggest that the project score has a stronger relationship with final exam score than the midterm score. Explain your answer.
 - (d) Interpret $\hat{\beta}_2$ in the context of the problem.
- (2) State if below statements are true or false with an explanation.
- (a) For MLR, the $Adj - R^2$ will always be smaller than R^2 value.

HINT: Two formulas can be be written as below

$$R^2 = 1 - \frac{SSE}{SSTotal}$$

$$Adj - R^2 = 1 - \frac{(n-1)}{(n-k-1)} \frac{SSE}{SSTotal}$$

- (b) If we fit a MLR model and then add a new predictor to the model, the $Adj - R^2$ will always increase.
- (3) You will be using *RailsTrails.csv* data for this activity. The primary goal of the researcher was to learn the impact on the sale price ($Adj2007$ in 1000's of dollars) by the distance to a bike trail. By adding the size of the house ($SquareFeet$ in 1000's) to the model, we hope to address this research question while controlling/adjusting for the size of the house.
- (a) Fit a SLR model of $Adj2007$ on $Distance$. What is the coefficient of determination for the model?
 - (b) Now, fit a model with both predictor variables: $Distance$ and $SquareFeet$. By how much R^2 has increased?
 - (c) Check the validity of the assumptions for the fitted model in part (b).
 - (d) When $Distance$ is in the model, does the contribution from $SquareFeet$ is significant?
 - (i) Set up appropriate null and alternative hypotheses.
 - (ii) Extract the value of the test statistic for the above test.

- (iii) What is the distribution of the test statistic under the null?
 - (iv) What is the p value for the test?
 - (v) Write the final conclusion for the above question (use 5% significance level)
 - (e) Using the two predictor model, obtain a prediction for the price of a particular home that goes in the market that has 1500 square feet and 0.5 miles from a bike trail. Provide an interpretation for this interval.
- (4) There are two types of dementia: Dementia with Lewy Bodies, and Alzheimer's disease. Some people are afflicted with both of these. You will be using *LewyBody2Groups* dataset in *Stat2Data* package for this analysis.
- (a) Use the `?LewyBody2Groups` in the console to see the help page for the dataset. Use it to provide the details of the variables in the dataset.
 - (b) Make a scatterplot of MMSE on APC with *Type* as a grouping variable (use different colors/shapes for the two types). Comment on your observations.
 - (c) Fit a model that produces parallel regression lines for the two levels of *Type*. Write down the fitted regression model. Define any terms in the model equation.
 - (d) What is the average estimated difference in MMSE for patients with both types of dementia compared to those with only Lewy Body dementia assuming they have similar percentage change from baseline volume of the brain?
 - (e) Is the average difference you stated in part (d) statistically significant (use 5% significance level)?
 - (i) Set up appropriate null and alternative hypotheses.
 - (ii) Extract the value of the test statistic for the above test.
 - (iii) What is the p value for the test?
 - (iv) Answer the reserach question based on the p value in part (iii)